

Project Initial Details Submission

1. Problem Statement and Target Users

1.1 What real-world problem does your application aim to solve?

Under the Singapore Green Plan 2030, commercial enterprises face mounting regulatory and supply chain pressures to decarbonize operations and account for Greenhouse Gas (GHG) Protocol emissions (National Climate Change Secretariat [NCCS], 2021; World Resources Institute & World Business Council for Sustainable Development [WRI/WBCSD], 2004). Although the Singapore Government aggressively co-funds private-sector transitions through schemes administered by Enterprise Singapore, such as the Energy Efficiency Grant (EEG), three in every four local Small and Medium-sized Enterprises (SMEs) remain unprepared to adopt sustainable practices due to capability constraints, with over 70% having never accessed available government assistance (Bain & Company, 2024; Enterprise Singapore [EnterpriseSG], n.d.; Goh, 2025).

This underutilization stems from a critical two-sided friction: on the applicant side, SMEs lack in-house carbon accounting expertise, leaving owners unable to translate practical equipment upgrades into statutory GHG Protocol boundaries such as Scope 1 direct fuel combustion, Scope 2 purchased grid electricity, or Scope 3 value chain (Energy Market Authority [EMA] & EnterpriseSG, 2024) or screen out disqualifying exclusionary conditions like second-hand or refurbished machinery (EnterpriseSG, n.d.); on the assessor side, public grant officers face multi-week backlogs because inbound proposals arrive as unstructured, free-form text narratives that require labor-intensive manual auditing to verify statutory criteria, calculate energy baselines, and detect prohibited clauses (EnterpriseSG, n.d.).

1.2 Who are the intended users of the application?

Primary Users (SME Applicants): SME business owners, operations managers, and facility directors seeking instant pre-assessment to verify eligibility, estimate subsidy caps, and eliminate compliance risks before formal submission.

Secondary Users (Grant Reviewers & Compliance Auditors): Public sector grant intake analysts who need an automated, high-speed triage pipeline to audit inbound proposals, isolate violations, and fast-track valid projects.

2. User Inputs

2.1 What information or data will users provide to the system?

The presentation layer (`io_manager.py`) collects structured metadata and an unstructured proposal via terminal prompts. The structured metadata will include: company name, annual revenue, employee count, percentage of equity owned by Singapore Citizens or Permanent Residents, which industrial sector category, baseline annual energy expenditure, estimated cost of retrofit, the sustainability grant the user is applying for.

The unstructured proposal narrative will include operational targets. (E.g., "Replacing our 6 diesel delivery vans with commercial electric vans and installing depot Level-2 smart chargers to eliminate Scope 1 fleet fuel combustion.").

3. Use of AI

3.1 How will AI be utilized within the application?

The AI processing layer (ai_manager.py) serves as the core semantic reasoning engine. Standard regex or keyword searches fail because they cannot discern context (e.g., a keyword search for "electric" would incorrectly pass "procuring second-hand electric vans", missing the prohibited "second-hand" condition).

3.2 What outputs, insights, or recommendations will the AI generate from the user inputs?

The AI will generate technical justifications of the emissions scope boundary, estimated abatement percentage, and operational bottlenecks along with predictions like the resulting energy usage reduction if a grant is taken and how to better align their business to best benefit from the available schemes as part of its recommendations. It will also recommend the most relevant grant scheme for the SME based on its business profile, proposed project, eligibility criteria, and potential benefits.

4. Business Rules

4.1 What business rules, validations, or decision-making logic will be applied to the AI-generated outputs?

The logic layer (logic_manager.py) evaluates the AI-generated JSON extractions alongside user financial data using deterministic, procedural Python rules to determine the application's grant eligibility, after which a terminal audit report is generated for the user. The heuristic rules will include: Statutory SME Eligibility Check, Negative Exclusion Screening, Tier 1 Fast Track Pre-Approval, Tier 2 Provisional Capability Approval, low-impact rejection, and persistence & advisory reporting.

5. Repository Information

GitHub URL: https://github.com/SYnthetic/INF1103_ProjectGrpWork.

Git Clone: https://github.com/SYnthetic/INF1103_ProjectGrpWork

References (APA 7th Edition)

- Bain & Company. (2024, October 3). *Overcoming barriers to Singapore SME decarbonisation*. <https://www.bain.com/insights/overcoming-barriers-to-singapore-sme-decarbonisation/>
- Energy Market Authority, & Enterprise Singapore. (2024, October). *Powering a sustainable future: A playbook on Scope 2 emissions reduction for Singapore businesses*. Government of Singapore. <https://www.ema.gov.sg/resources/industry-reports/playbook-scope-2-emissions-reduction-for-sg-businesses>
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- Goh, T. (2025, November 12). 3 in 4 SMEs not ready to go green due to tighter operating margins, lack of expertise: Report. *The Straits Times*.
<https://www.straitstimes.com/business/companies-markets/3-in-4-smes-not-ready-to-go-green-due-to-tighter-operating-margins-lack-of-expertise-report>
- National Climate Change Secretariat. (2021). *Singapore Green Plan 2030*. Government of Singapore. <https://www.greenplan.gov.sg>
- World Resources Institute, & World Business Council for Sustainable Development. (2004). *The Greenhouse Gas Protocol: A corporate accounting and reporting standard* (Rev. ed.). <https://ghgprotocol.org/corporate-standard>